Secure Remote Support Terminal for Android with Admin Dashboard and Command Execution

SENTRIOD

Project Proposal Document

PROJECT INFORMATION

Project Name:	Secure Remote Support Terminal for Android with Admin Dashboard and Command Execution				
Project Area:	Mobile Application Security, Android APK Assessment, DevSecOps, Agentic AI, Software Supply Chain Security		**End Users:**	Mobile pentesters, AppSec teams, third-party Android development companies, fintech/telco security teams, academic security labs	
Project Type:	~~✓ Research~~	✓ Development	**Technology:**	✓ Based	☐ Structured
Platform:	~~☐ Desktop~~	✓ Web App	~~☐ Mobile App~~	~~☐ Distributed~~	~~☐ Others~~
If Others:					

SUPERVISOR INFORMATION

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1. Executive Summary

This proposal presents SENTRIOD, a secure Mobile Device Management (MDM) solution specifically designed to support the operational, security, and administrative requirements of intelligence and law enforcement organizations. The solution aims to provide centralized control, continuous visibility, and policy-based management of official mobile devices used by personnel for communication, field operations, intelligence gathering, coordination, and access to sensitive organizational resources.

As intelligence organizations increasingly rely on mobile technologies to support mission-critical activities, the risks associated with unmanaged or inadequately protected devices have become significantly greater. Unauthorized access, data leakage, device compromise, loss or theft of operational devices, and non-compliance with security policies can directly impact operational effectiveness and national security interests. The absence of a centralized mobile device management capability also limits organizational visibility, accountability, and the ability to respond rapidly to security incidents.

SENTRIOD addresses these challenges by providing a unified and secure platform for device enrollment, policy enforcement, application management, real-time monitoring, remote administration, incident response, and comprehensive audit logging. The solution enables authorized administrators to maintain continuous oversight of deployed devices, enforce organizational security policies, manage device configurations, and rapidly respond to operational or security events from a centralized management console.

The proposed platform incorporates security-by-design principles, including role-based access controls, strong authentication mechanisms, encrypted communications, device compliance monitoring, and detailed audit trails. These capabilities support secure mobile operations while ensuring accountability, traceability, and adherence to organizational security requirements.

Designed for scalability and operational resilience, SENTRIOD can support deployment across headquarters, regional offices, and field units while maintaining centralized governance and control. The solution enhances situational awareness, strengthens cybersecurity posture, reduces operational risks associated with mobile device usage, and improves the efficiency of mobile asset management across the organization.

By implementing SENTRIOD, the organization will establish a secure and centrally governed mobile ecosystem that supports operational readiness, protects sensitive information, enhances device security, and enables effective management of mobile assets in support of intelligence and security operations.

2. Why Is Mobile Device Management Required?

The following operational challenges have been identified in mobile device usage:

2.1 Decentralized Usage

Mobile devices are currently administered separately by different departments or units, resulting in inconsistency in their configuration and security measures. This decentralized approach not only creates inconsistency in security controls but also makes it difficult to ensure all devices meet organizational standards. As a result, some devices may remain vulnerable due to weaker configurations or outdated settings. The absence of coordination also increases the overall attack surface for cyber threats. This decentralized management structure reduces visibility and control over mobile assets.

2.2 Policy Enforcement Gap

The current mobile device environment lacks a centralized mechanism for enforcing security policies, creating a significant policy enforcement gap. Security policies are not uniformly applied or monitored across all devices, leading to inconsistencies in configuration and compliance. Variations in operating systems and device types further widen this gap, as policies cannot be effectively standardized or enforced. As a result, some devices operate outside defined security controls, increasing the risk of vulnerabilities and non-compliance with organizational and regulatory requirements. This lack of consistent enforcement represents a critical gap in maintaining a secure and controlled mobile environment.

2.3 Delayed Incident Response

In the absence of centralized control, responding to security incidents involving mobile devices is often delayed. When devices are lost, stolen, or compromised, there is no immediate mechanism to remotely secure or wipe the devices. This delay increases the likelihood of unauthorized access to the sensitive data. Additionally, the lack of real-time monitoring makes it harder to detect and respond promptly to the real-time threats. Such delays can result in significant data breaches and operational risks.

2.4 Inefficient Manual Processes

Managing mobile devices manually requires significant time and effort from IT teams, especially in large organizations. These processes are often repetitive and prone to human error, which can lead to misconfigurations or missed updates. As the number of devices grows, manual management becomes increasingly difficult. This inefficiency due to manual processes can slow down the operations and reduce overall productivity. Thus, automating device management is essential to improve accuracy, efficiency, and scalability.

3. Proposed SENTRIOD Solution

The proposed Mobile Device Management (MDM) solution named SENTROID will provide an **end-to-end framework** for securing and managing mobile endpoints within IBD or any other organization.

3.1 Key Features of the Proposed Solution

The proposed solution will provide a unified, secure, and efficient framework for managing mobile devices across the organization, ensuring consistency, scalability, and enhanced control.

Following are the key features of the proposed SENTRIOD solution:

1. Centralized administrative dashboard.
2. Secure device enrollment and identification

3. Policy-based security enforcement
4. Remote device control without physical access
5. Real-time monitoring and alerts
6. Comprehensive administrative logging and reporting
7. Automated device management processes to reduce manual effort, minimize errors, and improve operational efficiency.

The SENTRIOD solution will be designed to be scalable, allowing the organization to efficiently manage an increasing number of devices without compromising performance or control. Its robust security architecture will ensure that all devices, data, and communications are protected through consistent policy enforcement and advanced safeguards. Additionally, the system will be made adaptable to evolving operational and technological requirements, making it suitable for integration with future tools, platforms, and security standards. This flexibility will enable the organization to respond effectively to the changing business needs and emerging cyber threats.

4. Project Objectives

The proposed SENTRIOD implementation aims to achieve the following objectives:

Objective	Description
Centralize Device Management	Establish a unified and centralized platform to manage, monitor, and control all organizational mobile devices, ensuring consistency, visibility, and administrative efficiency.
Enforce Security Policies Remotely	Ensure that security policies are applied and updated remotely across all of the devices, regardless of their location or operating system.
Enable Real-Time Monitoring	Provide continuous real-time visibility into device status, compliance levels, and potential security threats, enabling timely detection and response.

Reduce Administrative Effort	Automate key device management processes including enrollment, policy enforcement, monitoring, reporting, and incident response to minimize manual workload and improve operational efficiency.
Improve Device Security	Enhance the overall security posture by enforcing standardized security policies, preventing unauthorized access, and protecting sensitive data across all managed devices.

5. Comprehensive System Scope

The comprehensive system scope defines the key functional components of the proposed SENTRIOD system. These components operate within the overall workflow illustrated in **Error! Reference source not found.** and collectively enable secure device management, policy enforcement, monitoring, and administrative control throughout the device lifecycle.

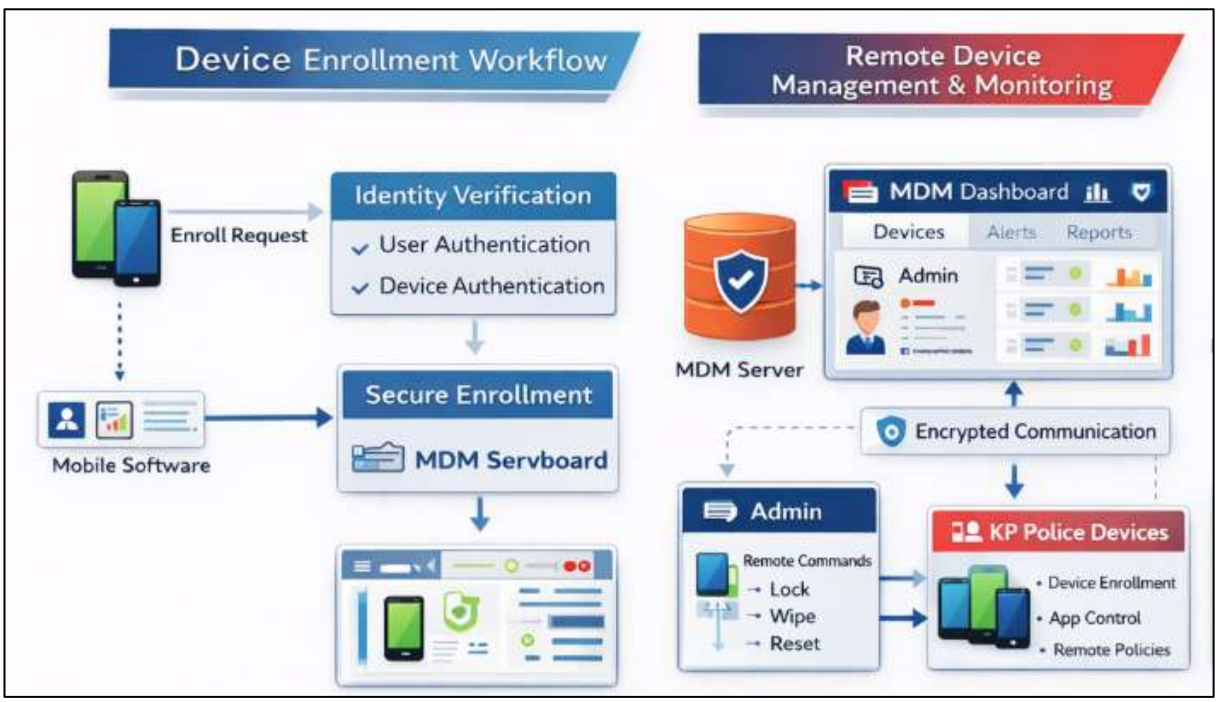

Figure 1 SENTRIOD System Workflow

Figure 1 illustrates the overall workflow of the proposed SENTRIOD system, including device enrollment, secure communication, administrative control, and monitoring processes. The functional components described in Sections 5.1 to 5.5 are logically represented within this workflow, although they are explained in detail in the following subsections for clarity. These components operate within the overall workflow illustrated in Figure 1 and represent the key functional modules of the SENTRIOD system, each corresponding to specific stages such as enrollment, control, policy enforcement, monitoring, and administration.

5.1 Device Enrollment & Registration

This phase focuses on the secure onboarding and proper registration of mobile devices within the system. It ensures that only authorized devices are allowed access and are configured according to organizational standards from the beginning. Proper enrollment establishes a solid foundation for device management and security compliance. The key features of this functionality are as follows:

- Android and iOS onboarding protection.
- Checking against organizational security standards.
- Delegation to users, departments, and units.

5.2 Remote Lock, Disable & Reset

This functionality enables administrators to take immediate action on devices that are lost, stolen, or compromised. It ensures that sensitive data remains protected even when devices are no longer physically accessible. Remote control capabilities significantly reduce the risk of unauthorized access. The key features of this functionality are as follows:

- Locking devices lost or stolen immediately.
- Turning on or off the use of a specific device remotely.
- Ability to wipe off data or to factory reset.

5.3 Policy Enforcement

This component ensures that all devices operate in accordance with predefined organizational security and usage policies. Policies are applied automatically, reducing the need for manual intervention and ensuring consistency across all devices. Continuous enforcement helps maintain compliance and strengthens overall security. The key features of this functionality are as follows:

- Security policies and usage policies are applied automatically.
- Constant adherence to the needs of the organization.

5.4 Device Status Monitoring

This feature provides ongoing visibility into the condition and compliance status of all managed devices. It enables administrators to proactively identify issues and respond to potential risks in a timely manner. Centralized monitoring improves decision-making and operational control. The key features of this functionality are as follows:

- Continuous monitoring of device health and policy compliance.
- Visibility Centralized on a single dashboard.

5.5 Administrative Control & Logging

This section ensures that administrative actions are properly controlled, tracked, and audited. It supports accountability by assigning roles and maintaining detailed records of system activities. Logging also assists in investigations and compliance reporting. The key features of this functionality are as follows:

- Role-based administrative control
- Detailed logs for auditing, accountability, and investigations

6. Core Functionalities

The proposed MDM solution will provide the following core functionalities:

1. **Device Enrollment & Registration:**

SENTROID will provide secure onboarding and identification of authorized mobile devices within the organization. It will ensure that all devices are validated against predefined security standards before being granted access to organizational resources.

2. **Remote Device Control:**

   SENTROID will enable remote control capabilities, including locking, wiping, disabling, or resetting devices in case of loss or compromise. This functionality will help protect sensitive data and prevent unauthorized access even when devices are not physically accessible.

3. **Policy Enforcement:**

   SENTROID will enforce centralized security and usage policies across all managed devices. It will ensure consistent compliance with organizational standards by automatically applying and updating policies as required.

4. **Device Monitoring & Alerts:**

   SENTROID will provide real-time visibility into device status, compliance levels, and potential security threats. It will also generate alerts for suspicious activities or policy violations, enabling timely response and risk mitigation.

5. **Administrative Control & Logging:**

   SENTROID will offer centralized administrative control with role-based access mechanisms. It will maintain detailed logs of all activities to support auditing, accountability, and security investigations.

6. **Centralized Management Dashboard:**

   SENTROID will provide a unified dashboard for monitoring, control, and reporting of all managed devices. It will enhance operational efficiency by giving administrators a comprehensive and centralized view of the entire device ecosystem.

7. Proposed System Architecture

The SENTRIOD solution will be implemented using layered and modular architecture.

7.1 Architecture Layers

The proposed SENTRIOD architecture shall support on-premises or hybrid deployment based on IBD data governance requirements. All device data, logs, and administrative controls shall remain under the ownership of IBD. The system can be integrated with existing authentication and IT infrastructure where required.

Layer	Function
Core Enrollment	Handles secure device authentication, certificate-based enrollment, and validation against organizational security policies before granting access.
Management & Control	Provides centralized device lifecycle management, including configuration, monitoring, and remote command execution through secure communication channels.
Policy Enforcement	Ensures continuous compliance by applying rule-based security policies, automated enforcement mechanisms, and real-time violation detection.

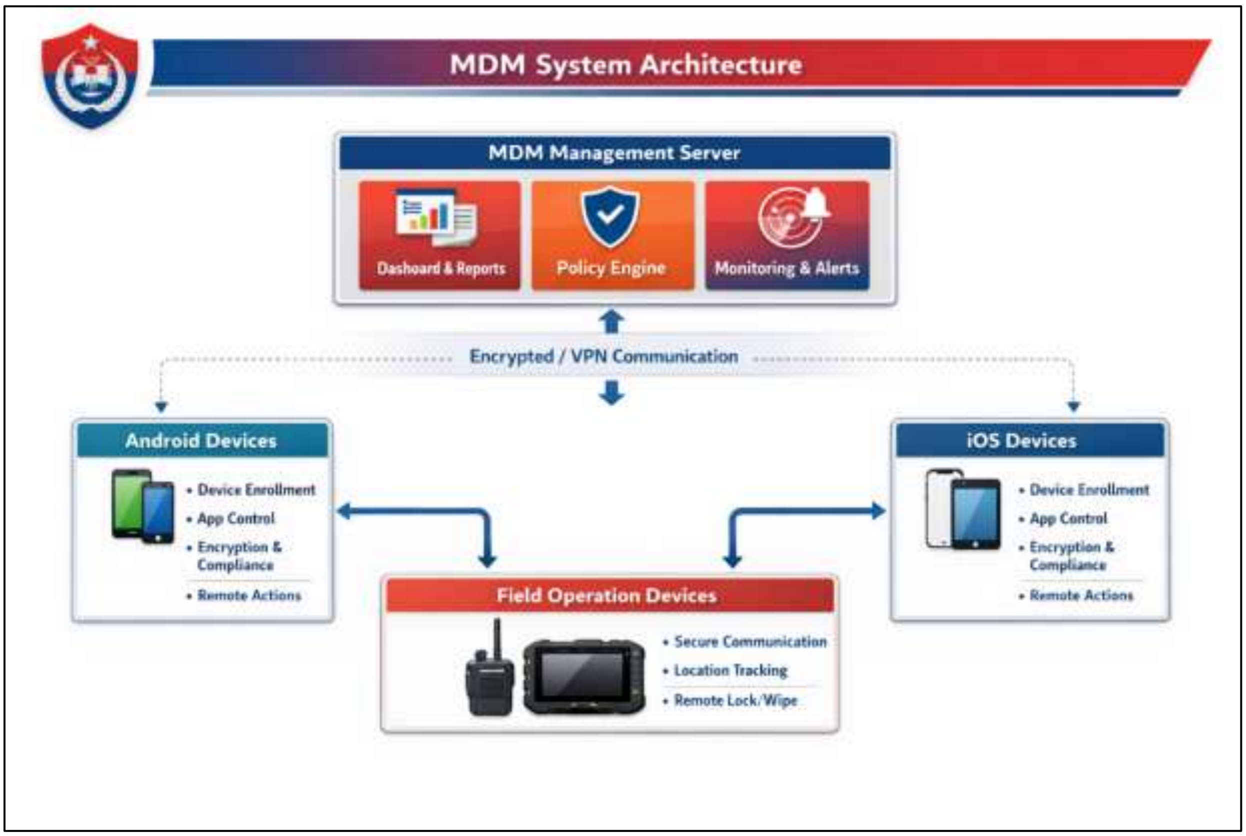

Figure 2 Proposed SENTRIOD System Architecture

Error! Reference source not found. illustrates the layered system architecture of the proposed SENTRIOD solution, highlighting the separation between device enrollment, management and control, and policy enforcement components. The architecture is designed to ensure secure device onboarding, centralized administrative control, and scalable policy application. Its modular structure supports future expansion and integration with existing IT and security systems.

7.2. Operational Workflow

The proposed operational workflow will include the following key processes (as shown in Figure 3):

1. Device Enrollment
2. Policy Assignment
3. Remote Command Execution
4. Device Response
5. Status Monitoring

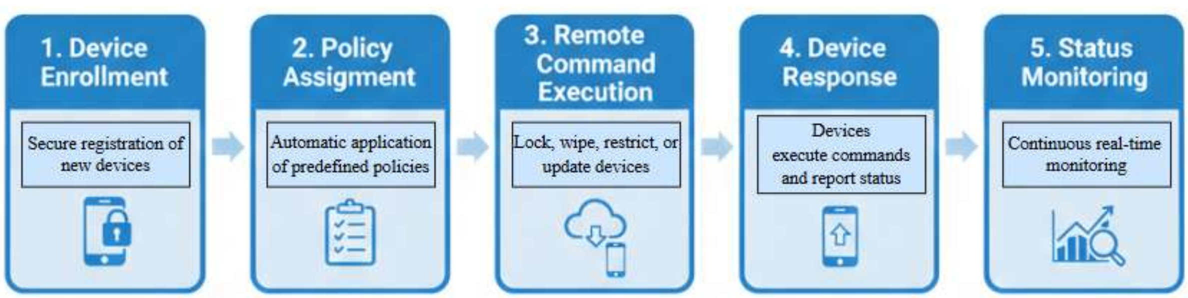

Figure 3 Operational Workflow

The operational workflow illustrates the interaction between administrators, the SENTRIOD platform, and managed mobile devices during routine operations and security events. Each stage in the workflow plays a critical role in ensuring secure, controlled, and efficient device management.

Given is the brief description of the key workflow processes:

- **Device Enrollment:** This stage involves the secure registration and authentication of devices into the SENTRIOD system. Devices are verified against predefined security requirements and configured to ensure compliance before being granted access.
- **Policy Assignment:** Once enrolled, predefined security and usage policies are automatically applied to devices. These policies ensure standardized configurations and continuous compliance with organizational security requirements.
- **Remote Command Execution:** Administrators can remotely issue commands such as lock, wipe, restrict, or update devices through the SENTRIOD platform. These commands are transmitted securely to ensure immediate response to operational needs or security incidents.
- **Device Response:** Managed devices execute the received commands and securely transmit execution status and system feedback back to the SENTRIOD server for validation, logging, and monitoring purposes.
- **Status Monitoring:** The system continuously monitors device activity, compliance status, and security events in real time. It generates alerts for anomalies or policy violations, enabling proactive and timely incident response.

7.3. Implementation Methodology

The implementation of the proposed SENTRIOD solution will follow a structured and phased approach to ensure controlled deployment, minimize risks, and achieve full operational readiness within the defined timeline. Each phase represents a critical step in ensuring a smooth and controlled deployment of the SENTRIOD system. The project will be executed in the following phases as shown in **Error! Reference source not found.**:

The implementation plan includes requirements analysis, system design, development, integration, testing, training, and final handover. The phased methodology ensures systematic development, effective risk management, timely delivery, and operational readiness. Each phase is aligned with the overall **120-day project timeline (1 May – 28 August)**. A brief description of each phase is provided below:

1. Requirements Analysis

- Detailed functional and security requirements are gathered from stakeholders.
- Use cases, operational workflows, and administrative requirements are analyzed.
- System constraints, performance expectations, and security objectives are identified.
- Requirements Specification Document (SRS) is prepared and approved.

2. System Design

- The overall solution architecture is designed, including Android agent, backend services, database, APIs, and web-based administration dashboard.
- Security architecture, authentication mechanisms, communication protocols, and access control models are defined.
- Database schema, system interfaces, and deployment architecture are finalized.
- UI/UX mockups and dashboard workflows are prepared.

3. Development

- The Android Support Agent is developed with device registration, secure communication, monitoring, and command execution capabilities.
- Backend services, APIs, authentication modules, and audit logging mechanisms are implemented.
- The web-based administration dashboard is developed for device management, monitoring, reporting, and command execution.
- Security controls and encryption mechanisms are integrated throughout the solution.

4. Integration and Configuration

- All system components are integrated and configured to operate as a unified platform.
- Communication channels between Android devices, backend services, and the dashboard are established.

- User roles, permissions, device registration workflows, and command authorization mechanisms are configured.
- Initial system validation is performed to verify interoperability.

5. Testing and Validation

- Functional testing is conducted to ensure all modules operate according to specifications.
- Security testing verifies authentication, authorization, encryption, and audit logging mechanisms.
- Performance and reliability testing validate system stability under operational conditions.
- Identified issues are resolved and the system is refined for production deployment.

6. Training, Deployment, and Handover

- System administrators and designated personnel receive training on platform operation, device management, monitoring, and reporting functions.
- Technical documentation, deployment guides, and user manuals are delivered.
- The completed system is deployed and formally handed over to the client along with source code, documentation, and maintenance recommendations.

This phased implementation approach ensures that the SENTROID solution is developed, tested, and deployed in a structured manner, resulting in a secure, reliable, and operationally effective platform within the 120-day project period.

7.4. Support and Maintenance

After the successful deployment of the SENTROID, ongoing support and maintenance are critical to ensure its optimal performance, adaptability, and sustainability. This phase focuses on providing technical assistance, monitoring system operations, refining policies, and building the capability of internal IT staff to manage the platform effectively. The key support and maintenance activities include:

- Basic post deployment technical assistance.
- Optimization and support of policies.
- Performance monitoring of the system.

***Scope Clarification**:*

The scope of this proposal excludes the procurement of end-user mobile devices and mobile data connectivity services.

7.5. Future Scalability and Adaptability

The proposed solution is designed to ensure future scalability and adaptability through the following capabilities:

- **Diversification to other devices and departments** – enabling the system to scale across the organization.
- **Interaction with future cybersecurity tools** – ensuring compatibility with evolving security measures.
- **Response to new versions of mobile operating systems** – maintaining seamless operation as technology updates.
- **Long-term operational sustainability** – providing a stable and maintainable framework for ongoing use.

8. Expected Outcomes and Benefits

Managing mobile devices in a law enforcement environment requires a systematic approach that balances operational efficiency with robust security. Implementing a Mobile Device Management (MDM) system like SENTRIOD will ensure that all devices are centrally controlled, monitored in real time, and comply with organizational policies. This structured management will not only streamline routine operations but will also strengthen the protection of sensitive intelligence data. The following table highlights the key benefits of SENTROID along with their expected impact:

Benefit	Expected Impact
Centralized Control	Establishes a unified management point for all mobile devices, simplifying administration and oversight.
Real-Time Response	Enables immediate detection and mitigation of security threats to prevent potential breaches.
Policy Adherence	Ensures consistent application of security policies across all devices, reducing compliance gaps.
Operational Efficiency	Minimizes manual intervention, saving time and resources while improving workflow.
Enhanced Security	Strengthens protection of sensitive law enforcement data against unauthorized access or leaks.

9. Timeline

Proposed timeline for the proposed solution spanning over 120 days is as follows:

SENTRIOD Implementation Plan – 120 Days Timeline

Project Start: 1st May | Project End: 28th August

PHASE	DURATION	TIMELINE	KEY DELIVERABLES
1 Literature Review & Requirement Analysis	30 DAYS	1st May – 30th May	• Literature review • Requirement document
2 System Design	15 DAYS	31st May – 14th June	• System architecture • Policy framework
3 Development	25 DAYS	15th June – 9th July	• Android app development • Admin dashboard development • Command execution module
4 Deployment & Configuration	15 DAYS	10th July – 24th July	• Platform deployment • System configuration
5 Enrollment & Policy Application	15 DAYS	25th July – 8th August	• Device enrollment • Policies applied
6 Testing & Validation	10 DAYS	9th August – 18th August	• Functional testing • Security validation
7 Training & Documentation	10 DAYS	19th August – 28th August	• Training completion • User manuals & documentation

TOTAL DURATION: 120 DAYS | PROJECT START: 1st May | PROJECT END: 28th August

10. Conclusion

This proposal presents a comprehensive and secure Mobile Device Management (MDM) **solution SENTRIOD** tailored to the operational needs of the **IBD**. By enabling centralized control, real-time monitoring, and policy-driven security enforcement, the system will significantly enhance mobile governance and reduce cybersecurity risks.

With a scalable architecture and structured implementation approach, the proposed solution will ensure long-term sustainability, operational efficiency, and improved protection of sensitive law enforcement data. The development and deployment of this system will be a strategic step toward modernizing mobile infrastructure and strengthening digital security within the organization.